Practice Questions – CAT II Preparation

1. A nighttime security camera captures images containing noise and weak object boundaries. Compare Canny, LoG, and DoG for detecting edges. Select the most suitable method and justify your choice in terms of noise handling and localization accuracy.
2. A robot captures images of a warehouse containing shelves, boxes and straight structural boundaries. Design a feature-extraction pipeline using an edge detector, Hough Transform and a corner detector. Explain the purpose of each stage and the expected output after every stage.
3. A mobile application has to recognize the same product package when it appears at different sizes, rotations, and viewing angles. Compare Harris and Hessian-Affine for detecting interest points and SIFT/SURF and HOG/GLOH for describing them. Select a suitable combination and explain how scale-space or image pyramids can improve recognition.
4. A thermal image of an industrial component contains a defective region with intensity variations and moderate noise. A few pixels inside the defective area are known. Explain how Region Growing can be used to segment the defective region. Discuss the role of seed selection, similarity threshold, and neighbourhood connectivity.
5. An agricultural vision system has to extract a diseased region from a leaf image containing noise, non-uniform intensity, and weak boundaries**.** Propose an interactive hybrid segmentation approach incorporating Region Growing, edge information, and Graph-Cut**.** Describe the order in which you would apply these techniques and justify the purpose of each. Explain how user-selected seeds or scribbles can guide segmentation and how the final region can be refined for subsequent disease/object detection**.**
6. A satellite image contains a flooded region with noise, intensity variations, and partially weak boundaries. Design a hybrid segmentation pipeline using Region Growing, an edge-based method, and Graph-Cut. Explain the sequence of these methods and the role of each stage. Discuss how seed points can be used and

how the combined approach can reduce region leakage and incorrect segmentation.

7. A manufacturing plant collects high-dimensional sensor measurements to identify abnormal machine behaviour. Only a few abnormal samples are labelled.

   a) Explain how K-Medoids or GMM can be used to discover natural groups in the sensor data.

   b) Select an appropriate supervised or semi-supervised classifier from Bayes, KNN, or ANN for identifying abnormal patterns. Justify your selection considering the limited labelled data.

   c) Explain how a discriminant function can be used to decide whether a new sensor sample belongs to a known group or should be flagged as anomalous. Also discuss the effect of outliers.

8. A video-analysis application has to process high-dimensional visual features and then track moving objects in a fixed-camera scene**.** Select a suitable dimensionality-reduction technique from PCA, LDA and ICA and combine it with a non-parametric approach**.** The output is then processed using Background Subtraction and Optical Flow**.**

   Explain the complete sequence, the purpose of each technique and how reducing the feature dimensions may affect computational efficiency and discriminative information. Also explain how motion information can be estimated from consecutive frames.

9. A surveillance system extracts a large number of image features and subsequently tracks moving people.

   a) Select PCA, LDA, or ICA for reducing the feature dimensions and explain its purpose.

b) Explain how a non-parametric method such as KNN can operate on the reduced feature vectors. How does dimensionality reduction affect its computational cost?

c) Explain how Background Subtraction and Optical Flow can be used together to detect moving people and estimate their direction and motion.

10. An industrial inspection system must detect scratches and textured defects whose sizes and orientations vary between images.

    a) Explain how Scale-Space Analysis using Image Pyramids can help detect defects of different sizes. What is the role of Gaussian smoothing?

    b) Explain how Gabor Filters and DWT can represent oriented and multi-resolution texture information.

    c) Add one method from SIFT, SURF, or HOG to the pipeline and explain its contribution. Compare the resulting hybrid approach with using only an edge or line detector, and mention its computational limitation

Problem

Select technique

Draw pipeline

Explain each stage

Give mathematical/basic principle

Justify selection

Mention advantage

Mention limitation